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Turkish Question - Answer Dataset Evaluated with Deep Learning

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***Abstract*—The field of Natural Language Processing (NLP) poses various challenges, one of which is the task of question-answering which involves comprehension, multiple-choice, yes-no questions, and more. Unfortunately, creating question-answer dataset is one of the main obstacles to these studies, especially in some languages such as Turkish where dataset availability is limited. To address this scarcity, this study developed a Turkish question-answering dataset from the comprehensive English SQuAD dataset using Microsoft Translator. We initially checked all translator output and observed that we can use approximately 60% of data after updating index of answer. Then 16% of all data was refined in Turkish language for increasing data amount. As a result, we successfully created 78% of the SQuAD in Turkish language. Additionally, a question-answering deep learning model was built to test the dataset's usability and model performance was assessed using various metrics, domain-specific considerations, translation quality, and linguistic statistics.**

Keywords*—Natural Language Process, Question Answering, Machine Reading Comprehension, Turkish Dataset, SQuAD

I. Introduction

Turkish language is structurally distinct from English language due to its agglutinative nature. A sentence in English can be expressed with only one word in Turkish language, as demonstrated in Figure 1. These structural distinctions may necessitate modification and adaptation.

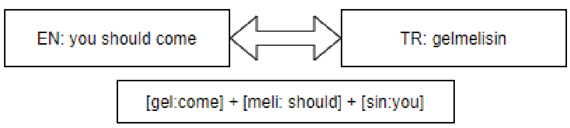

Fig. 1. Example of structural differences between Turkish and English.

Question-answering tasks are a challenging set of tasks in NLP, one of them is Machine Reading Comprehension (MRC). MRC tasks aim to find answers in specific documents and might focus on open or closed domains. SQuAD, published by [1] is a comprehensive open-domain MRC dataset in English. Various question-answering models have been developed to achieve high accuracy in SQuAD, especially those based on deep learning models. Recently, transformer architectures have been built with attention mechanism that handles long-range context dependencies. Also, they provide powerful language representations, Large Language Model (LLM), that enable understanding the context in detail.

This study translates the SQuAD dataset into Turkish using Microsoft Translator [2]. After the translation process, the rate of directly usable data is approximately 60% and remain data has different errors. Fixed and refined translation error rate was calculated to 16% for the entire dataset. After all the work, usable Turkish data rate was increased to 78% by using different fixed and cleaning methods, which will be explained in the next sections. Gemirter [3] previously translated SQuAD into Turkish with 50% of the data availability. To evaluate the usability of the translation approach, the dataset of [3] was used as a control dataset. In addition to dataset creation, two question-answering models were developed and trained using BERT infrastructure [4]. One model was trained on the new created dataset in this study, called m-SQuAD (output of the Microsoft translator). The other model was trained on a dataset of [3], called g-SQuAD (output of the Google translator). The results were analyzed from different perspectives, such as the importance of the data context on the model outputs, the usability of the translation approach to create dataset, fixed and refined processes after the translation and the importance of linguistic issues on model behaviors.

II. Related work

There are different types of question-answering datasets, especially for the English language. The most important one of these is the SQuAD dataset [1]. Developing such a resource for a new language requires a lot of effort. In this study, the SQuAD v1.1 is translated into Turkish language by using machine translation. SQuAD v1.1 is a large-scale machine reading comprehension dataset; it contains +100,000 questions, crowd-sourced from Wikipedia articles. The training dataset consists of 442 titles, while the test dataset includes 48 titles. Titles contain many paragraphs, each paragraph contains many questions and a question can have more than one answer.

In Turkish, the THQuAD [5] dataset has started to be developed in recent years. It has the same structure as the SQuAD dataset, but it is not as rich in content as SQuAD at the moment. Still, it is an important study to create a new open-domain question-answering dataset for Turkish language. This dataset has 15554 questions unfortunately only two different domains Ottoman History and Turkish Islamic Science history. The richness of content is important in question-answering systems because the models may achieve high success in general topics (country, city, biography of famous people) but they may not show the same performance in more detailed topics such as philosophy and science. The study [3] in Turkish, the SQuAD 1.1 dataset is translated into Turkish with 50% data loss using Google Translator API. In our study, 78% of SQuAD 1.1 is translated into Turkish by using Microsoft Translator and performing fixed operations after the translation process. The dataset of [3] is used as the control dataset to evaluate the translation and model performances. Another study [6] focuses only on creating the dataset phase of the question-answering system. All questions of applied official exams in Türkiye are collected and compiled for research studies but its data size is so limited. However, the size of training and test data is critical in AI-based question answer systems.

Besides there are similar studies in different languages. [7] is created by SQuAD translating into Spanish. In this, they used a custom infrastructure for the translation process. After the translation, they applied some heuristics to clean and refine the retrieved (context, question, answer) examples and they have two datasets which are SQuAD-es v1.1, which is a bigger but noisy dataset, SQuADes-small (53% of the original data) that is the smaller but more accurate. Another study is [8] to French. They translated a subset of SQuAD into French and extracted several paragraphs and their corresponding question/answer pairs, resulting in 327 (question, answer, paragraph pairs) over 48 articles so the results are very limited.

Neural network models have contributed significantly to the performance of question-answering tasks with their language presentations and strong attention mechanisms. [9] presented a mechanism that some of the inputs should be more important than others and model supports that giving more importance to them. In 2018, Bidirectional Encoder Representations from Transformers (BERT) [4] was published by Google. One of the most important models in this field has achieved a significant accuracy rate on the SQuAD dataset. According to the state of art table of SQuAD [14], BERT has EM(Exact Match): 87% F1: 93% accuracy in English. Also, different neural network approaches with pre-trained language representations are available, some of them are ELMo [15] and Generative Pre-trained Transformer (GPT) [10]. ELMo provides a bidirectional architecture however adaption of different tasks is harder than other solutions. GPT required fewer architectural differences also it was unidirectional. [3] is one of the important question-answering study in Turkish. They fine-tuned the BERT model on different question-answering datasets, also the study has been used different language models BERTurk [13] and custom LLMs. They achieved EM:%55 and F1:%67 scores on a limited SQuAD dataset. Another recent study [5] has models developed on the THQuAD dataset (similar to SQuAD to Turkish but so limited). They fine-tuned some neural network models and their results are ELECTRA EM:%63 F1:%81, BERT EM:%62 F1:%81, ALBERT EM:%48, F1:%69. BERT includes a comprehensive presentation of language. BERTurk [13] is one of the studies carried out for modeling the Turkish language model on BERT. In this study, BERTurk is used as the large language model (LLM).

Another study, rule-based system for question analysis in Turkish, [11]. The study focuses on analyzing the question part of the question and answer system. After analyzing the question, it creates a focus information from the question. The found focus information may be used to search for answers on different data sources. Generalizing rule-based studies and feature extraction approaches for open domains require a lot of human power and language-specific studies.

III. Dataset

A. SQuAD Dataset

The Stanford Question Answering Dataset (SQuAD) consists of sets of content (c), questions (q), and answers (a) triples. The dataset was compiled from 10,000 Wikipedia articles and is divided into two categories: training and development sets. The training set comprises 90% of the articles, while the development set contains 10% of the articles. Compared to other important machine reading comprehension datasets, such as MCTest, which only has 2,640 questions, SQuAD is larger and more challenging, containing numerous questions that require direct answers from a single passage. This structure makes the task more difficult for NLP models. While some document-based question-answering datasets focus on answer extraction by finding related sentences or paragraphs, SQuAD emphasizes finding the correct answer as one or more specific words. Statistics and sample data for SQuAD can be found in Tables I and II.

TABLE I
The statistics of SQuAD

Statistic Items	Number-of-Items and Value
Question	Train:87599 Development:10570
Paragraph	Train:18896 Development:2067
Title	Train:442 Development:48
Human EM Score	86.831
Human F1 Score	89.452
IE-Net EM (Sota) Score	90.939
IE-Net F1 (Sota) Score	93.214

TABLE II
The sample data of SQuAD

Item	Value
Title	Nikola Tesla
Paragraph	In 1870, Tesla moved to Karlovac, to attend school at the Higher Real Gymnasium, where he was profoundly influenced by a math teacher Martin Sekuli0̆107.:32 The classes were held in German, as it was a school within the Austro-Hungarian Military Frontier...
Question	Who was Tesla influenced by while in school?
Answer1	Martin Sekuli
Answer2	Martin Sekuli
Answer3	math teacher Martin Sekuli
Answer Start Point	148

B. The Creation of mSQuAD

The training and development datasets of SQuAD were translated into Turkish language using Microsoft Translator and published on GitHub [12]. The translation process took 3 days. Since Microsoft charges per character for each request, the translation was completed with a budget of approximately $400. As a result, the rate at which ”content contains answer” was calculated to be 62%. This rate means that about half of the data may not be usable. Additionally, there is another dataset, referred to as gSQuAD, which was translated using Google Translator and contains approximately 50% of SQuAD [3]. gSQuAD was used to compare and assess the datasets. To evaluate the effect of data quantity and work with a larger dataset, we performed many different fixes operations to increase our usable output of translation. This approach also helps to understand how the quality of translation affects accuracy of model. For this purpose, the content (c), question (q), and answer (a) structures were translated in batches using Microsoft Translator. After identifying and correcting various translation errors, we increased the dataset's usable size to 78%. Sample data from the translated dataset is shown in Table III.

TABLE III
THE SAMPLE DATA OF MSQUAD

Item	Value
Title	Nikola Tesla
Paragraph	1870'te Tesla, Yüksek Gerçek Spor Salonu'ndaki okula gitmek için Karlovac'a taşındı ve burada matematik öğretmeni Martin Sekulić'den derinden etkilendi. Dersler Avusturya-Macaristan Askeri Sınırı ...
Question	Tesla'nın Karlovac'taki ana etkisi kimdi?
Answer1	Martin Sekulić
Answer2	Martin Sekulić
Answer3	matematik öğretmeni Martin Sekulić
Answer Start Point	128

C. Post Translation Actions

As a result of the post-translation analysis, we realized approximately 60% of the data was included in correct triple form (content(c)-answer(a)-questions(q)) for model training and testing. Microsoft Translator provided %10 more initial usable data than Google Translator. To reduce data loss, error cases were examined and the loss data was included in the usable dataset with data fix batch codes on translated data.

TABLE IV
EFFECT OF THE POST-TRANSLATION CORRECTION TYPE TO TRANSLATED ANSWERS

Case-Number	Devevelopment-Dataset-Rate	Train-Dataset-Rate
Case 1	+%5	+%3
Case 2	+%6	+%5
Case 3	+%4	+%4
Case 4	+%1	+%1

Case 1: Basic syntax errors were observed in some answers from translation outputs. There were dot, comma, and space characters at the end of the translation (ex: en: "alternating current", tr: "alternatif akım.") also there were similar characters with different encoding values in the translation outputs (ex: '–,-', content has:"14-12" answer has :"14-12"). Moreover, there were a lot of date format translation errors, such as en-answer: "February 7, 2016" tr-answer after translation: "7 Şubat 2016, Şubat 2016, Bugün", correct answer: "7 Şubat 2016".

Case 2: Since answers appeared with different casings in the context, all data was converted to lowercase. However, this operation was not sufficient for all matching tasks. Turkish documents may contain specific English terms and also exhibit language-specific casing details (e.g., Turkish: I-ı, I-İ, i-i versus English: I-i). We found exact matches between the context and the answers after adjusting for casing and replacing language-specific characters. Finally, we addressed casing issues using custom-developed methods for uncapitalizing and capitalizing.

Case 3: Some English proper nouns in the answers were translated into Turkish. In such cases, if the original English answer was present in the Turkish content, the translated answer was updated to match the original English answer. For example, "Arizona Cardinals," a sports team in the USA, appears as "Arizona Cardinals" in the Turkish content, but the translation was given as "Arizona Kardinalleri." in answer part. Additionally, when English answers contain the word "The" it was removed from the original English answer data during comparison. These adjustments contributed extra data to the dataset.

Case 4: There are some discrepancies between the content and answers due to differences in the translation of monetary values, numbers, and mathematical notations. For example, the content might contain "1,200" while the answer translates to "1.200," or "10%" in the content might be translated as "%10" in the answer. Similarly, "5,000,000$" in the content might be contained as "5.000.000 dollar" in the answer. These discrepancies were corrected to ensure consistency between the context and answers. In the batch script, these characters or terms were either removed from the translation output or replaced with the correct values. The impact of these corrections on the dataset size is shown in Table IV.

D. Statistics of mSQuAD

With the help of changing the automatic translation service and fixing batch codes to translation errors, a significant increase in usable data was achieved. Compared to the previous study [3] as gSQuAD, 28% more data is available, and these statistics are given in Table V for the development and train dataset.

TABLE V
THE TRANSLATION OUTPUT STATISTIC OF MSQUAD

Statistic	Dev	Train
SQuAD Total q-a pair	34726	87599
Raw a-in-c after Translation	21530	54311
Edited q-a pair	5796	11323
Usable q-a pair	27326	65634
Increased Rate with Post-Translation Process	+%16	+%12
Increased Rate than gSQuAD	+%27	+%25
SQuAD Total Usable q-a pair Rate	%78	%75

When we investigated why we could not import the remaining data into the dataset, we observed required human correction. There were different errors in many different situations with reasons as given in Table VI.

TABLE VI
NEEDED HUMAN OPERATIONS AFTER TRANSLATION

Answer-En	Answer-Tr	Content-Tr	Type
pressure terms	basınç koşulları	...basınç terimlerinin...	in-consistency
second	saniye	...ikinci sezon...	synonymy
linebacker	Linebacker	...defans oyuncusu...	not translated
the metric slug	metrik sümüklüböğürtlen	metrik sümüklüböcek	wrong translated
broken arm	kırık kol	...kırık bir kolla...	language differences
grass	çimen	...çimin bir kısmı...	language differences

IV. MODEL

A. Bidirectional Encoder Representations from Transformers (BERT)

In recent years, transformer-based models have achieved significant success on question-answering tasks due to their strong architectures and large language presentations. BERT is one of the most important of them. When BERT was proposed, it achieved state-of-the-art accuracy on many NLP tasks (such as language understanding and question answering) [4]. BERT runs with a semi-supervised approach to learning. Firstly the model is trained to understand the patterns of the language structure with a different large corpus as Wikipedia. Then fine-tune process on BERT provides task-specific learning. BERT is basically an encoder-decoder network stack of transformer architecture that uses self-attention. This architectural structure enables a better understanding and interpretation of the context.

B. Fine-Tuned Model

Since BERT language model training is a long and tiring process, the pre-trained BERTURK [13] is used as the Turkish large language model (LLM). The current version of BERTURK is trained on a filtered and sentence-segmented version of the Turkish OSCAR corpus, a recent Wikipedia dump, various OPUS corpora, and a special corpus. Its corpus has a size of 35GB and 4,404,976,662 tokens. Using mSQuAD-train-tr and gSQuAD-train-tr dataset (as given their statistics in Table VII), m-QA-Model-Tr and g-QA-Model-Tr models were fine-tuned with BERTURK as shown in the Figure 2. Fine-tune process took about 30 hours with BERTBASE structure, PyTorch library.

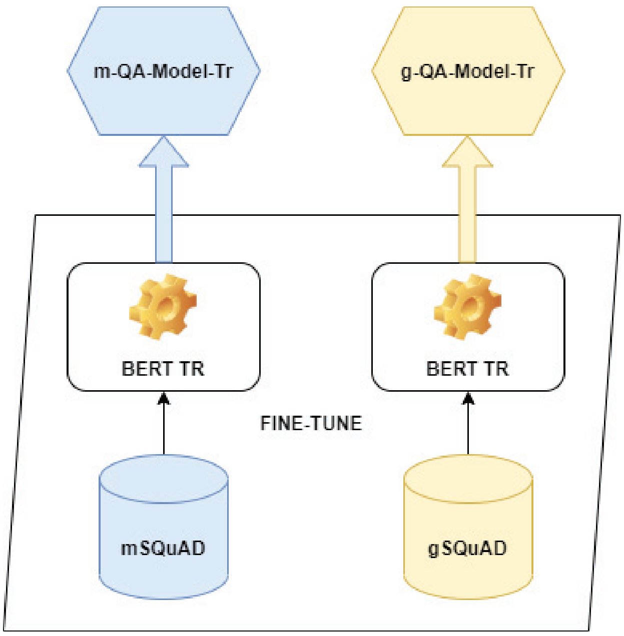

Fig. 2. Trained two QA models (m-QA-Model-Tr, g-QA-Model-Tr) based on mSQuAD and gSQuAD

TABLE VII
SQuAD, mSQuAD and gSQuAD Dataset Question/Title/Paragraph Count Statistics.

Dataset	Question	Title	Paragraph
SQuAD-dev-en	10570	42	2067
mSQuAD-dev-tr	8197	42	2067
gSQuAD-dev-tr	5858	42	2067
SQuAD-train-en	87599	448	18896
mSQuAD-train-tr	65634	448	18896
gSQuAD-train-tr	40014	448	18896

V. Results

A. Evaluation

The m-QA-Model-Tr and g-QA-Model-Tr were evaluated using various test cases. First, both models were tested on the entire mSQuAD-dev-tr dataset. Then, they were evaluated on common questions found in both the gSQuAD and mSQuAD datasets, first on mSQuAD-dev-tr and then on gSQuAD-dev-tr. Additionally, the models were assessed from different perspectives on common questions found in mSQuAD-dev-tr, including question type (e.g., who, when, what), BLEU score (translation quality), token size of c-q-a, and paragraph titles. Our analysis showed that the model trained on mSQuAD outperformed better EM and F1 scores, showing a difference of 1% to 2%. For more detailed information, refer to Table VIII. We also experimented with a different approach by replacing the questions that the m-QA-Model-Tr failed to answer with questions from the gSQuAD-dev-tr dataset, and vice versa. This approach resulted in a significant improvement in results, highlighting the importance of both the questions and content. Additionally, we observed that increasing the amount of data positively impacted the success of the models.

TABLE VIII
m-QA-Model-Tr and g-QA-Model-Tr models results

Model	EM	F1	Test Dataset
m-QA-Model-Tr	0.49	0.57	mSQuAD dev-tr
g-QA-Model-Tr	0.47	0.55	mSQuAD dev-tr
m-QA-Model-Tr	0.52	0.59	mSQuAD-dev-tr (only common questions)
g-QA-Model-Tr	0.50	0.58	mSQuAD-dev-tr (only common questions)
m-QA-Model-Tr	0.50	0.57	gSQuAD-dev-tr (only common questions)
g-QA-Model-Tr	0.48	0.56	gSQuAD-dev-tr (only common questions)
m-QA-Model-Tr	0.56	0.63	Question-Change at fails
g-QA-Model-Tr	0.55	0.62	Question-Change at fails

By analyzing the impact of the length of content (c), questions (q), and answers (a) on the model's behavior, we found that longer inputs have a negative effect on the success of the model. Additionally, there is significant correlation between input length and stability. We calculated the accuracy values based on the average token limits for each input type, and the results are presented in Table IX.

TABLE IX
m-QA-Model-Tr results on mSQuAD-dev-tr with tokenize count perspective as c =content, q= question, a= answer

Model	EM	F1	Token Count	Type
m-QA-Model-Tr	0.54	0.62	≤ 95	c
m-QA-Model-Tr	0.45	0.51	≥ 95	c
m-QA-Model-Tr	0.52	0.60	≤ 7	q
m-QA-Model-Tr	0.49	0.56	≥ 7	q
m-QA-Model-Tr	0.54	0.60	$= 1$	a
m-QA-Model-Tr	0.43	0.52	$= 2$	a
m-QA-Model-Tr	0.37	0.47	$= 3$	a

It's crucial to note that a paragraph's content is associated with a specific domain. Generally, models perform better with general content like countries, cities, and biographies, but we observed more challenges with scientific terminology and fields. The details of these findings are presented in Table X and XI.

The accuracy of the models was greatly influenced by the context of the questions being asked. While the models performed well with simple questions that have clear answers,

TABLE X
m-QA-Model-Tr results on mSQuAD-dev-tr with domain oriented, top-5, bottom-5

Domain	EM	F1	EM Rank
Islamism	0,72	0,77	1
1973 oil crisis	0.68	0.74	2
Kenya	0.66	0.76	3
Harvard University	0.63	0.63	4
Sky (United Kingdom)	0.62	0.66	5
Civil disobedience	0.38	0.47	44
European Union law	0.36	0.53	45
Prime number	0.35	0.43	46
Chloroplast	0.34	0.48	47
Computational theory	0.34	0.51	48

TABLE XI
g-QA-Model-Tr results on mSQuAD-dev-tr with domain oriented, top-5, bottom-5

Domain	EM	F1	EM Rank
United Methodist Church	0,72	0,74	1
1973 oil crisis	0.68	0.76	2
Islamism	0.67	0.76	3
Kenya	0.64	0.76	4
Nikola Tesla	0.63	0.70	5
Immune system	0.36	0.47	44
Ctenophora	0.35	0.47	45
Oxygen	0.34	0.46	46
Packet switching	0.32	0.41	47
Computational theory	0.30	0.43	48

such as ”when” ”how much” and ”who”. They struggled with more open-ended questions like ”which” and ”what”. Detailed results for these different types of questions can be found in Tables XII and XIII.

TABLE XII
m-QA-Model-Tr results on mSQuAD-dev-tr with question type, q type count ≥ 100

Question Type	EM	F1	EM Rank	Count
Ne zaman (When)	0,76	0,80	1	423
Kaç (How much)	0.69	0.73	2	443
Ne kadar (How much)	0.62	0.67	3	117
Kim	0.57	0.63	4	570
Nerede (Where)	0.45	0.52	5	151
Hangi (Which)	0.51	0.59	6	996
Ne (What)	0.42	0.52	7	1661

TABLE XIII
g-QA-Model-Tr results on mSQuAD-dev-tr with question type, q type count ≥ 100

Question Type	EM	F1	EM Rank	Count
Ne zaman (When)	0,72	0,75	1	423
Kaç (How much)	0.69	0.72	2	443
Ne kadar (How much)	0.57	0.64	3	117
Kim	0.56	0.62	4	570
Nerede (Where)	0.53	0.62	5	151
Hangi (Which)	0.52	0.59	6	996
Ne (What)	0.40	0.50	7	1661

To assess the quality of translations, we used the consistent gQuAD-dev-tr data and evaluated the second mSQuAD-dev-tr set by calculating the BLEU score. We found that a lower BLEU score indicates decreased accuracy, these results can be found in Table XIV.

TABLE XIV
m-QA-Model-Tr results on mSQuAD-dev-tr with tokenize count perspective as c =content, q= question, a= answer

Model	EM	F1	Bleu	Type
m-QA-Model-Tr	0.54	0.61	≥ 0.75	c
m-QA-Model-Tr	0.45	0.51	≤ 0.75	c
m-QA-Model-Tr	0.54	0.61	≥ 0.75	q
m-QA-Model-Tr	0.47	0.54	≤ 0.75	q
m-QA-Model-Tr	0.53	0.60	$= 1$	a
m-QA-Model-Tr	0.40	0.51	≤ 1	a

VI. Conclusions

Our study focused on enhancing the dataset and improving the accuracy of our data generation method. Additionally, our research demonstrates the effectiveness of machine translation in facilitating NLP tasks across various languages. We also developed deep learning models to observe how different aspects of the data affect the model's performance. By identifying the challenges that deep learning models face with Turkish data, we have highlighted areas that require further research and development for future studies. Our goal in future work is to achieve balanced model performance across the entire dataset by conducting detailed research on long contexts (content, questions, answers), question types, and domains where the model currently struggles.

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